Jun Xiao

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Education

Ph.D., The Hong Kong Polytechnic University *Hong Kong*

Full scholarship, Department of Electronic and Information Engineering, Supervised by Prof. Kin-Man Lam *09/2018 - 09/2022*

- **Thesis topic:** Machine Learning for Image Super-resolution in Real-world Applications

Master, The Hong Kong Polytechnic University *Hong Kong*

*MSc in Electronic and Information Engineering **(Distinction)**, Supervised by Prof. Kin-Man Lam* *09/2016 - 09/2018*

- **Award**: The Outstanding Post-graduate Student Scholarship. (**Top 5%**)

Bachelor, Guangdong University of Technology *Guangzhou, China*

BEng in Telecommunication Engineering *09/2012 - 06/2016*

- **Award**: The Outstanding Student Scholarships. (**Top 5%**)

Working Experience

Video AI Algorithm Engineer, Zoom Video Communications *Singapore*

Full-time, video processing team (Core member, research and model development) *14/11/2024-present*

- **Responsibility**: Text-to-image/video generation, Generative-based low-level vision processing, High-quality SFT dataset curation,
- **Project 1**: Logo-driven image generation. 1). the subject consistency score achieve 0.855; 2). Generation time is within 2s for 1280 x 720 resolution.
- **Project 2**: Text-driven style transfer for human images. 1). Human ID consistency is 0.85; 2). Style consistency is 0.63. 3). The generation time is within 1s for 512x512 resolution.

Postdoctoral Fellow, The Hong Kong Polytechnic University *Hong Kong*

Full-time, Grant by Research Talent Hub, Supervised by Prof. Kin-Man Lam *10/202-10/2024*

- **Research project**: Lead the project "efficient old movie 4K photorealistic restoration": design an efficient video restoration model for low-resolution old movies without ground truth.
- **Results**: 1). Improve 4x processing speed. 2). the accepted rate achieve about 85%. 3). deliver Cinema-released Movie, link: Nomad.

Computer Vision Researcher, Microsoft Research Asia (MSRA) *Shanghai, China*

Internship, mentors: Xinyang Jiang, Ningxin Zheng, and Dongsheng Li *09/2021-01/2022*

- **Research project**: Involved in the project "online video restoration and enhancement system": design efficient and lightweight super-resolution model for online streaming videos;
- **Results**: 1). On 2080Ti GPU, the inference speed achieves 110 FPS for 720p videos. 2). Model parameters are only 1.75M; 3). Publish in IEEE Transactions on Multimedia.
- **Award**: MSRA Stars of Tomorrow **(Award for Excellent Intern)**

Selected Publications

1. Cuixin Yang, Rongkang Dong, **Jun Xiao**, et al. "Geometric distortion guided transformer for omnidirectional image super-resolution." **IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)**, 2025.
2. **Jun Xiao**, Zihang Lyu, Cong Zhang, Yakun Ju, Changjian Shui, and Kin-Man Lam. "Towards Progressive Multi-Frequency Representation for Image Warping", in Proceedings of the **IEEE Conference on Computer Vision and Pattern Recognition (CVPR)**, 2024.
3. **Jun Xiao**, Changjian Shui, Zhi-Song Liu, Qian Ye, and Kin-Man Lam, "Learning Equilibrium Transformation for Gamut Expansion and Color Restoration", in Proceedings of the **European Conference on Computer Vision (ECCV)**, 2024.
4. Kin-Chung Chan, **Jun Xiao**, Hana Lebeta Goshu, and Kin-Man Lam. "Point Cloud Densification for 3D Gaussian Splatting from Sparse Input Views", in Proceedings of the **ACM Conference on Multimedia (ACM-MM)**, 2024. (**Corresponding Author**)
5. **Jun Xiao**, Qian Ye, Tianshan Liu, Cong Zhang, and Kin-Man Lam, "Deep Progressive Feature Aggregation Network for Multi-frame High Dynamic Range Imaging", **Neurocomputing**, 2024.
6. **Jun Xiao**, Qian Ye, Rui Zhao, Kin-Man Lam, and Kao Wan, "Deep Multi-scale Feature Mixture Model for Image Super-resolution with Multiple-Focal-length Degradation", **Signal Processing: Image Communication**, 2024.

7. **Jun Xiao**, Zihang Lyu, Hao Xie, Cong Zhang, Yakun Ju, Changjian Shui, and Kin-Man Lam. "Frequency-Aware Guidance for Blind Image Restoration via Diffusion Models", in Proceedings of the **European Conference on Computer Vision Workshop (ECCV-W)**, 2024.
8. Yushen Zuo, **Jun Xiao**, Kin-Chung Chan, Rongkang Dong, Cuixin Yang, Zongqi HE, Hao Xie, and Kin-Man Lam. "Towards Multi-View Consistent Style Transfer with One-Step Diffusion via Vision Conditioning", in Proceedings of the **European Conference on Computer Vision Workshop (ECCV-W)**, 2024. (**Corresponding Author**)
9. Zhi-Song Liu, Li-Wen Wang, **Jun Xiao**, *et al*. "Bridging Text and Image for Artist Style Transfer via Contrastive Learning", in Proceedings of the **European Conference on Computer Vision Workshop (ECCV-W)**, 2024.
10. **Jun Xiao**, Kin-Man Lam, et al. "Online Video Super-Resolution with Convolutional Kernel Bypass Graft", **IEEE Transaction on Multimedia (TMM)**, 2023.
11. **Jun, Xiao**, Qian, Ye, Rui Zhao, Kin-Man Lam, and Kao Wan. "Self-feature Learning: An Efficient Deep Lightweight Model for Image Super-resolution", in Proceedings of the **ACM Conference on Multimedia (ACM-MM)**, 2021.
12. Qian Ye, **Jun Xiao**, et al. "Progressive and Selective Fusion Network for High Dynamic Range Imaging", in Proceedings of the **ACM Conference on Multimedia (ACM-MM)**, 2021. (**Equal Contribution**)
13. **Jun Xiao**, Tianshan Liu, Rui Zhao, and Kin-Man Lam, "Balanced Distortion and Perception in Single-Image Super-Resolution Based on Optimal Transport in Wavelet Domain", **Neurocomputing**, 2021.
14. **Jun Xiao**, Rui zhao, Kin-Man Lam, et al, "Bayesian Sparse Hierarchical Model for Image Denoisin", **Signal Processing: Image Communication**, 2021.
15. **Jun Xiao**, Wenqi Jia, and Kin-Man Lam, "Feature Redundancy Mining: Deep Light-weight Image Super-resolution Model", in the **International Conference on Acoustics, Speech and Signal Processing (ICASSP)**, 2021.
16. **Jun Xiao**, Rui Zhao, Shun-Cheung Lai, Wenqi Jia, and Kin-Man Lam, "Deep Progressive Convolutional Neural Network for Blind Super-Resolution With Multiple Degradations", in Proceedings of the **IEEE International Conference on Image Processing (ICIP)**, 2019.
17. Rui Zhao, Tianshan Liu, **Jun Xiao**, et al. "Invertible Image Decolorization", **IEEE Transactions on Image Processing (TIP)**, 2021.

Academic Experience

- **Regular Reviewer**:
  - Journals: IEEE-TCSVT; IEEE-TMM; IEEE-TIP; Pattern Recognition; Knowledge-based Systems; Information Fusion; Neurocomputing
  - Conferences: NeurlPS, ICLR, CVPR, ECCV, ICCV, ACM-MM, WACV, ICIP, ICASSP.

Professional Skills

Programming Skills	Python (proficient), Matlab, C++, Pytorch (proficient), Microsoft Azure (proficient), Spark, Scikit-Learn (proficient), SQL.
Languages	Mandarin (native speaker), Cantonese (native speaker), English